

ASE 203 · Mathematical Methods for Engineering

Complex Numbers, Fourier & Laplace

複素数・フーリエ・ラプラス

Complete, worked study notes — in four parts

Part I · Complex Numbers & Phasors ■ Part II · Fourier Series
Part III · The Fourier Transform ■ Part IV · The Laplace Transform

after the lectures of **Prof. Faouzi Lakrad**

references: Kreyszig, *Advanced Engineering Mathematics*; Boyce & DiPrima, *Elementary Differential Equations and Boundary Value Problems* (Wiley, 2017)

How to read these notes

These notes rebuild both lecture decks from the ground up: every definition, property, theorem and worked example is here, with the algebra carried out in full and short *intuition* notes added wherever the idea behind a formula is worth stating in plain words. A handful of short sections also add standard companion material — division of complex numbers, phasors and AC impedance, half-range sine and cosine series, operational properties, and a one-page summary of the three Fourier forms — so the two topics stand complete on their own. The colour of the bar on the left of a box tells you what kind of thing it is:

Definition	a term being introduced and pinned down precisely.
Theorem	a central result you can rely on and quote.
Worked Example	a problem solved step by step, exactly as in lecture.
Intuition	the "why" — the mental picture behind the mathematics.
Properties	a list of rules that follow from a definition.

A pink-bordered **key box** highlights a formula worth memorising.

1 The course at a glance

This module builds the mathematical toolkit for *solving models written as differential equations* and for *analysing signals*. The whole map fits together like this:

- ♦ A single **ordinary differential equation (ODE)** is attacked with the **Laplace transform**.
- ♦ A **system of ODEs** is attacked with **matrices** (eigenvalues, modal decomposition).
- ♦ **Signal analysis and processing** is built on **Fourier series and the Fourier transform**.

ODEs are the language of mathematical modelling — they describe the dynamics of particles and rigid bodies, electric circuits, population dynamics in biology, chemical kinetics, and even arise from the reduction of partial differential equations (PDEs).

INTUITION the "solvable territory"

ODEs split into **linear** and **nonlinear**. Linear ODEs are the comfortable, solvable territory; nonlinear ones are not solvable in general, and we often *linearise* them (a Taylor expansion about an operating point) to bring them back into that territory. A second split is by side-conditions: an **initial value problem (IVP)** fixes the state at one instant, while a **boundary value problem (BVP)** fixes values at the ends of an interval. Fourier series live naturally with periodicity and BVPs; Laplace and Fourier transforms are the integral-transform machinery, all of the general shape

$$\mathcal{T}(f(t))(s) = \int_{\alpha}^{\beta} k(s, t) f(t) dt,$$

where $k(s, t)$ is the kernel that defines the particular transform.

The eight lectures of ASE203:

#	Topic	What it covers
1	Complex Numbers & Phasors	Complex plane, Euler's identity, polar form, phasors for oscillations and AC circuits.
2	Fourier Series	Orthogonality, coefficient computation, energy interpretation, engineering examples.
3	Fourier Transform	From series to transform, spectra, filtering, signals and vibration analysis.
4	Laplace Transform	Definition, properties, transform of derivatives, the meaning of $s = \sigma + i\omega$.
5	Solving ODEs with Laplace	Transfer functions, step response, damping, RLC and mass-spring-damper systems.
6	Eigenvalues & Eigenvectors	Diagonalisation, natural modes, stability intuition, engineering applications.
7	Systems of Linear ODEs	Matrix exponentials, modal decomposition, coupled engineering systems.
8	Applications & Synthesis	Vibration analysis, filtering, dynamic response, integrating the transform methods.

SCOPE OF THESE NOTES

These notes cover **Lecture 1 (Complex Numbers)** as Part I, **Lecture 2 (Fourier Series)** as Part II, **Lecture 3 (Fourier Transform)** as Part III, and **Lecture 4 (Laplace Transform)** as Part IV — closing with solving ODEs by Laplace (the start of Lecture 5). The remaining lectures are listed above only to place this material in context.

PART I

Complex Numbers 複素数



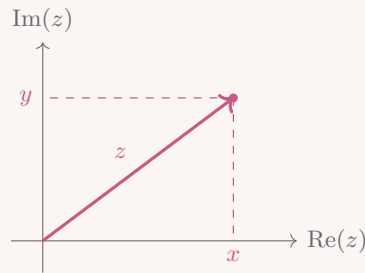
2 Cartesian form and the complex plane

A **real** number lives on a single line. A **complex** number is a two-dimensional number — an ordered pair of real numbers — and so it lives on a *plane*.

DEFINITION Complex number, Cartesian form

$$z = x + iy, \quad x, y \in \mathbb{R}, \quad i^2 = -1, \quad z \in \mathbb{C}.$$

The horizontal axis carries the **real part** $\operatorname{Re}(z) = x$; the vertical axis carries the **imaginary part** $\operatorname{Im}(z) = y$. The plane on which z is plotted is the **complex plane**.



For two numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, the algebra is governed by:

ALGEBRAIC PROPERTIES

- ♦ **Commutativity of addition:** $z_1 + z_2 = z_2 + z_1 = (x_1 + x_2) + i(y_1 + y_2)$.
- ♦ **Associativity of addition:** $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$.
- ♦ **Commutativity of multiplication:** $z_1 z_2 = z_2 z_1 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$.
- ♦ **Associativity of multiplication:** $z_1(z_2 z_3) = (z_1 z_2) z_3$.
- ♦ **Distributivity:** $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$.
- ♦ **Equality:** $z_1 = z_2 \iff x_1 = x_2 \text{ and } y_1 = y_2$.

INTUITION where the product rule comes from

Multiply $(x_1 + iy_1)(x_2 + iy_2)$ and use $i^2 = -1$: the real part is $x_1 x_2 - y_1 y_2$ and the imaginary part is $x_1 y_2 + x_2 y_1$. Equality of two complex numbers is *two* real equations at once — this is exactly the trick that powers Worked Example 1.

THE CAYLEY–DICKSON TOWER

Complex numbers are the second rung of a ladder of number systems, each doubling the dimension and giving up one algebraic comfort:

$$\underbrace{\mathbb{R}}_{1\text{-D}} \longrightarrow \underbrace{\mathbb{C}}_{2\text{-D, } i^2=-1} \longrightarrow \underbrace{\mathbb{H}}_{4\text{-D, } i^2=j^2=k^2=-1} \longrightarrow \underbrace{\mathbb{O}}_{8\text{-D}} \longrightarrow \underbrace{\mathbb{S}}_{16\text{-D}}.$$

$\mathbb{R}$ real, $\mathbb{C}$ complex, $\mathbb{H}$ quaternions, $\mathbb{O}$ octonions, $\mathbb{S}$ sedenions.

WORKED EXAMPLE 1 · find all z with $z^2 = i$

Write $z = a + ib$ with $a, b \in \mathbb{R}$. Then

$$z^2 = (a + ib)^2 = (a^2 - b^2) + i2ab.$$

Matching real and imaginary parts of $z^2 = i$ gives two real equations:

$$\begin{cases} a^2 - b^2 = 0 & (1) \\ 2ab = 1 & (2) \end{cases}$$

From (2), $b = \frac{1}{2a}$. Substituting into (1),

$$a^2 - \frac{1}{4a^2} = 0 \implies \frac{4a^4 - 1}{4a^2} = 0 \implies 4a^4 = 1 \implies a^4 = \frac{1}{4}.$$

Since a is real we keep $a^2 = \frac{1}{2}$ (the branch $a^2 = -\frac{1}{2}$ has no real solution):

$$a = \frac{1}{\sqrt{2}} \Rightarrow b = \frac{1}{\sqrt{2}} \Rightarrow z = \frac{1+i}{\sqrt{2}}, \quad a = -\frac{1}{\sqrt{2}} \Rightarrow b = -\frac{1}{\sqrt{2}} \Rightarrow z = \frac{-1-i}{\sqrt{2}}.$$

So $z^2 = i$ has the two square roots $z = \pm \frac{1+i}{\sqrt{2}}$.

3 The complex conjugate

DEFINITION Conjugate

$$\bar{z} = x - iy.$$

Geometrically $\bar{z}$ is the **mirror image** of z across the real axis.

CONJUGATE RULES

- ♦ $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ and $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$.
- ♦ $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$ ($z_2 \neq 0$) and $\bar{\bar{z}} = z$.
- ♦ Recover the parts: $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$, $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$.

4 Modulus

DEFINITION Modulus / absolute value

$$|z| = \sqrt{x^2 + y^2}.$$

Because $z\bar{z} = (x + iy)(x - iy) = x^2 + y^2$, we also have the clean identity

$$\boxed{|z| = \sqrt{z\bar{z}}} \quad \text{equivalently} \quad z\bar{z} = |z|^2.$$

5 Division, and properties of the modulus

Division is handled with the conjugate: multiply numerator and denominator by $\bar{z}_2$ to make the denominator real.

DEFINITION Quotient of two complex numbers

$$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{z_1 \bar{z}_2}{|z_2|^2} \quad (z_2 \neq 0).$$

WORKED EXAMPLE · a quotient in Cartesian form

$$\frac{3 + 2i}{1 - i} = \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)} = \frac{3 + 5i + 2i^2}{1^2 + 1^2} = \frac{1 + 5i}{2} = \frac{1}{2} + \frac{5}{2}i.$$

MODULUS

- ♦ $|z| \geq 0$, and $|z| = 0 \iff z = 0$; $|\bar{z}| = |z|$.
- ♦ $|z_1 z_2| = |z_1| |z_2|$, $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$, $|z^n| = |z|^n$.
- ♦ $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$.
- ♦ **Triangle inequality:** $|z_1 + z_2| \leq |z_1| + |z_2|$, with reverse form $||z_1| - |z_2|| \leq |z_1 - z_2|$.

INTUITION lengths multiply, the triangle still holds

Because multiplication multiplies lengths and adds angles, the modulus behaves like a genuine distance from the origin — including the triangle inequality, the statement that a detour through a third point is never shorter than going straight there.

6 Polar form and the argument

Reading the right triangle in the complex plane with hypotenuse $r = |z|$ and angle θ from the positive real axis gives the polar form.

DEFINITION Polar form

$$z = r e^{i\theta} = r(\cos \theta + i \sin \theta), \quad \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

with $r = |z|$ and

$$\theta = \arctan\left(\frac{y}{x}\right) \equiv \arg(z).$$

The value of $\arg(z)$ lying in $-\pi < \theta \leq \pi$ is the **principal value**, written $\text{Arg}(z)$, and

$$\arg(z) = \text{Arg}(z) + 2k\pi, \quad k \in \mathbb{Z}.$$

INTUITION $\arctan$ is not the whole story

The bare formula $\theta = \arctan(y/x)$ only knows the *ratio* y/x , so it cannot tell the second quadrant from the fourth, or distinguish $+1$ from -1 . You must place the point in the correct quadrant by eye. That is exactly why $\text{Arg}(-1) = \pi$ even though $\arctan(0/(-1)) = 0$ — see case 3 below.

WORKED EXAMPLE 2 · polar form and principal argument

Write each number in polar form and give its principal argument.

$$1) z = i \quad 2) z = 1 \quad 3) z = -1 \quad 4) z = 1 - i \quad 5) z = -i$$

1. $z = 0 + 1i$: $|z| = 1$, $\theta = \arctan(1/0) = \frac{\pi}{2} + 2k\pi$, so $\text{Arg}(i) = \frac{\pi}{2}$.
2. $z = 1 + 0i$: $|z| = 1$, $\theta = \arctan(0/1) = 0$, so $\text{Arg}(1) = 0$.
3. $z = -1 + 0i$: $|z| = 1$. Here $\arctan(0/(-1)) = 0$, *but* the point sits on the negative real axis, so $\text{Arg}(-1) = \pi$.
4. $z = 1 - 1i$: $|z| = \sqrt{2}$, $\theta = \arctan(-1/1) = -\frac{\pi}{4}$, so $\text{Arg}(1 - i) = -\frac{\pi}{4}$.
5. $z = 0 - 1i$: $|z| = 1$, $\theta = \arctan(-1/0) = -\frac{\pi}{2}$, so $\text{Arg}(-i) = -\frac{\pi}{2}$.

7 Euler's formula and De Moivre

The polar form rests on **Euler's formula**, the bridge between the exponential and the trigonometric world:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

PROPERTIES OF $e^{i\theta}$

- ◆ $|e^{i\theta}| = 1$ and $\overline{e^{i\theta}} = e^{-i\theta}$ and $e^{-i\theta} = \frac{1}{e^{i\theta}}$.
- ◆ $e^{i(\theta+\varphi)} = e^{i\theta} e^{i\varphi}$ and $(e^{i\theta})^n = e^{in\theta}$.
- ◆ $\arg(\bar{z}) = -\arg(z)$, $\arg(\frac{1}{z}) = -\arg(z)$, $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$.

THEOREM De Moivre's formula

$$(\cos \theta + i \sin \theta)^n = (e^{i\theta})^n = e^{in\theta} = \cos(n\theta) + i \sin(n\theta).$$

INTUITION multiply by adding angles

In polar form, multiplication *adds angles and multiplies lengths*. Raising to a power therefore *multiplies the angle* and raises the length to that power — which is why polar form is the right tool for large exponents.

$$e^{i\pi} + 1 = 0 \quad (\text{Euler's identity})$$

Solving Euler's formula for the trigonometric functions gives the **inverse Euler relations** — the workhorses of Part II:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

WORKED EXAMPLE 3 · show $(\sqrt{3} + i)^3 = 8i$ two ways

Cartesian (binomial expansion).

$$(\sqrt{3} + i)^3 = 3\sqrt{3} + 9i - 3\sqrt{3} - i = 8i.$$

Polar. For $z = \sqrt{3} + i$ the modulus is $r = 2$ and $\theta = \arctan(\frac{1}{\sqrt{3}}) = \frac{\pi}{6}$, so $z = 2e^{i\pi/6}$ and

$$z^3 = 2^3 e^{i\pi/2} = 8i.$$

Which method for $(\sqrt{3} + i)^{30}$? The polar one — De Moivre turns it into $2^{30} e^{i30\pi/6} = 2^{30} e^{i5\pi}$ in one line, whereas the binomial expansion would be hopeless.

WORKED EXAMPLE · a negative power: $(\sqrt{3} + i)^{-6}$

With $z = 2e^{i\pi/6}$,

$$(\sqrt{3} + i)^{-6} = 2^{-6} e^{-i6\pi/6} = \frac{1}{64} e^{-i\pi} = -\frac{1}{64}.$$

WORKED EXAMPLE · a large power via polar: $(1 + i)^{10}$

Write $1 + i = \sqrt{2} e^{i\pi/4}$. Then

$$(1 + i)^{10} = (\sqrt{2})^{10} e^{i10\pi/4} = 2^5 e^{i5\pi/2} = 32 e^{i\pi/2} = 32i.$$

The Cartesian route would mean expanding a tenth power — polar form makes it a one-liner.

8 Roots of complex numbers

First, equality in polar form: if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then

$$z_1 = z_2 \implies \begin{cases} r_1 = r_2 \\ \theta_1 = \theta_2 + 2k\pi. \end{cases}$$

This $+2k\pi$ freedom is what produces *several* roots.

THEOREM The n roots of $z^n = w$

Write $z = r e^{i\theta}$ and $w = \rho e^{i\varphi}$. Then

$$z^n = r^n e^{in\theta} = \rho e^{i\varphi} \implies \begin{cases} r^n = \rho \\ n\theta = \varphi + 2k\pi \end{cases} \implies \begin{cases} r = \sqrt[n]{\rho} \\ \theta = \frac{\varphi}{n} + \frac{2k\pi}{n} \end{cases}$$

so that

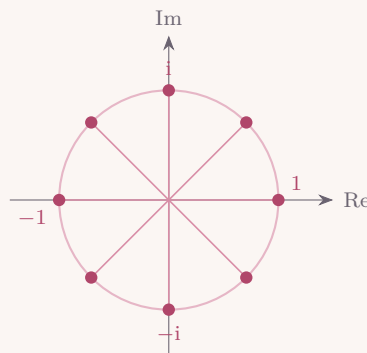
$$z = w^{1/n} = \sqrt[n]{\rho} e^{i(\frac{\varphi}{n} + \frac{2k\pi}{n})}, \quad k = 0, 1, \dots, n-1.$$

There are exactly n distinct roots, equally spaced by $2\pi/n$ around a circle of radius $\sqrt[n]{\rho}$.

WORKED EXAMPLE 4 · solve $z^8 = 1$

Write $z = r e^{i\theta}$, so $z^8 = r^8 e^{i8\theta}$, and $1 = e^{i2\pi k}$. Hence $r = 1$ and $\theta = \frac{\pi}{4}k$ for $k = 0, 1, \dots, 7$, i.e. $z = e^{ik\pi/4}$. Using $e^{i\theta} = \cos \theta + i \sin \theta$, the eight roots of unity are

$$\begin{aligned} k=0: & 1 & k=1: & \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} & k=2: & i & k=3: & -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \\ k=4: & -1 & k=5: & -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} & k=6: & -i & k=7: & \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \end{aligned}$$



The eight solutions of $z^8 = 1$: vertices of a regular octagon on the unit circle, spaced by $2\pi/8$.

WORKED EXAMPLE 5 · evaluate $\sqrt{4i}$

Here $w = 4i = 4e^{i\pi/2}$ and $n = 2$, so

$$w^{1/2} = 2e^{i(\frac{\pi}{4} + \frac{2k\pi}{2})}, \quad k = 0, 1,$$

giving

$$\sqrt{4i} = \begin{cases} 2e^{i\pi/4} = \sqrt{2} + i\sqrt{2} \\ 2e^{i5\pi/4} = -\sqrt{2} - i\sqrt{2}. \end{cases}$$

WORKED EXAMPLE 6 · evaluate $\sqrt[3]{-8}$

Here $w = -8 = 8e^{i\pi}$ and $n = 3$, so $w^{1/3} = 2e^{i(\frac{\pi}{3} + \frac{2k\pi}{3})}$, $k = 0, 1, 2$:

$$\sqrt[3]{-8} = \begin{cases} 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) = 1 + i\sqrt{3} \\ 2(\cos \pi + i \sin \pi) = -2 \\ 2(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}) = 1 - i\sqrt{3}. \end{cases}$$

Note that the familiar real answer -2 is just *one* of the three cube roots.

9 Phasors: complex numbers in motion

The reason complex numbers dominate engineering is that they turn the *calculus* of sinusoids into *algebra*. A real sinusoid of fixed frequency ω is encoded by a single complex number — its **phasor**.

DEFINITION Phasor

The sinusoid $x(t) = A \cos(\omega t + \phi)$ is represented by the complex **phasor**

$$X = A e^{i\phi} \quad \text{so that} \quad x(t) = \operatorname{Re}\{X e^{i\omega t}\}.$$

Here $A = |X|$ is the amplitude and $\phi = \arg(X)$ the phase.

WHY PHASORS WORK

- ♦ **Same frequency adds like vectors:** two sinusoids at the same ω add by adding their phasors — a hard trig sum becomes one complex addition.
- ♦ **Differentiation $\rightarrow$ multiplication:** $\frac{d}{dt} \rightarrow i\omega$. Each derivative multiplies the phasor by $i\omega$ (a 90° rotation and a scaling by ω).
- ♦ **Integration $\rightarrow$ division:** $\int dt \rightarrow \frac{1}{i\omega}$.

INTUITION a spinning arrow and its shadow

Picture $Xe^{i\omega t}$ as an arrow of length A spinning anticlockwise at rate ω , starting at angle ϕ . The real signal $x(t)$ is its shadow on the real axis. Since every term in a linear circuit spins at the same rate, you may freeze the rotation and work only with the stationary arrows X .

WORKED EXAMPLE · adding two sinusoids

Add $x_1(t) = 3 \cos(\omega t)$ and $x_2(t) = 4 \cos(\omega t + \frac{\pi}{2})$. In phasors,

$$X = 3e^{i0} + 4e^{i\pi/2} = 3 + 4i = 5e^{i0.927},$$

so $x_1(t) + x_2(t) = 5 \cos(\omega t + 0.927)$ — one sinusoid of amplitude 5 and phase $\approx 53.1^\circ$. The 3–4–5 right triangle does all the work.

AC CIRCUITS — THE PAYOFF

In a circuit driven at frequency ω , each element has a complex **impedance** Z , and Ohm's law becomes $V = ZI$ between phasors:

$$Z_R = R, \quad Z_L = i\omega L, \quad Z_C = \frac{1}{i\omega C}.$$

Series and parallel rules then look exactly like resistors, but with complex numbers — the bridge into the Laplace transform in Lectures 4–5.

PART II**Fourier Series** フーリエ級数**10 Periodic functions****DEFINITION** Periodic function

A function (signal) $f(t)$ is **periodic** with minimal period $T \neq 0$ (the **fundamental period**) if

$$f(t + T) = f(t) \quad \text{for all } t.$$

Every integer multiple of the fundamental period is also a period:

$$f(t + nT) = f(t), \quad n \in \mathbb{Z}.$$

WORKED EXAMPLE 1 · periods of the standard signals

$\cos(\omega t)$ and $\sin(\omega t)$ are periodic with fundamental period $T = \frac{2\pi}{\omega}$.

Is $f(t) = e^{i\omega t}$ periodic? Require $f(t) = f(t + T)$:

$$e^{i\omega t} = e^{i\omega(t+T)} = e^{i\omega t} e^{i\omega T} \implies e^{i\omega T} = 1 \implies \omega T = 2\pi k, \quad k \in \mathbb{Z}.$$

The smallest positive solution gives the fundamental period $T = \frac{2\pi}{\omega}$.

INTUITION a period is a repeating texture

A periodic signal carries one *pattern* (a texture) over a window of width T , and then repeats that exact pattern forever in both directions. Everything in Fourier analysis is about decomposing that one pattern into pure sines and cosines.

INTEGRATING PERIODIC FUNCTIONS

Let f be periodic with period $T > 0$ and integrable on every finite interval. Then the integral over *any* full period is the same:

$$\int_{\alpha}^{\alpha+T} f(t) dt = \int_0^T f(t) dt \quad \text{for every } \alpha \in \mathbb{R}.$$

In particular, taking $\alpha = -T/2$ centres the window:

$$\int_{-T/2}^{T/2} f(t) dt = \int_0^T f(t) dt.$$

Symmetry then gives two shortcuts that we use constantly:

$$f \text{ even } (f(-t) = f(t)) \implies \int_{-T/2}^{T/2} f dt = 2 \int_0^{T/2} f dt,$$

$$f \text{ odd } (f(-t) = -f(t)) \implies \int_{-T/2}^{T/2} f dt = 0.$$

DEFINITION Frequency, and its two readings

$$\omega = \frac{2\pi}{T} \quad (\text{angular / circular frequency}), \quad \text{freq} = \frac{1}{T}.$$

The units depend on what t measures:

- ♦ **t is time [s]:** T is the temporal period [s], ω is the angular frequency [rad/s], freq is the frequency [Hz].
- ♦ **t is a spatial coordinate [m]:** T is the wavelength λ [m], ω is the wavenumber [rad/m], freq is the spatial frequency [1/m].

11 Orthogonality

DEFINITION Scalar (inner) product; orthogonality

Two functions $f(t)$ and $g(t)$ are **orthogonal** on $[\alpha, \beta]$ if

$$\int_{\alpha}^{\beta} f(t) \overline{g(t)} dt = 0,$$

where the overline denotes the complex conjugate. A set of functions is **mutually orthogonal** if every distinct pair in the set is orthogonal.

WORKED EXAMPLE 2 · the trigonometric system is orthogonal

The set $\{1, \sin(n\omega t), \cos(n\omega t)\}$, $n = 1, 2, \dots$ is mutually orthogonal on $[-\frac{T}{2}, \frac{T}{2}]$ (with $T = 2\pi/\omega$). The full table of inner products is

$$\begin{aligned} \int_{-T/2}^{T/2} \sin(n\omega t) \cos(m\omega t) dt &= 0, & \int_{-T/2}^{T/2} \cos(n\omega t) dt &= \int_{-T/2}^{T/2} \sin(n\omega t) dt = 0, \\ \int_{-T/2}^{T/2} \sin(n\omega t) \sin(m\omega t) dt &= \int_{-T/2}^{T/2} \cos(n\omega t) \cos(m\omega t) dt = \begin{cases} T/2, & m = n \\ 0, & m \neq n. \end{cases} \end{aligned}$$

Proof for the sine pair, using $\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b))$.

Case $m \neq n$:

$$\begin{aligned} \int_{-T/2}^{T/2} \sin(n\omega t) \sin(m\omega t) dt &= \frac{1}{2} \int_{-T/2}^{T/2} \cos((n-m)\omega t) dt - \frac{1}{2} \int_{-T/2}^{T/2} \cos((n+m)\omega t) dt \\ &= \frac{1}{2} \left[\frac{\sin((n-m)\omega t)}{(n-m)\omega} - \frac{\sin((n+m)\omega t)}{(n+m)\omega} \right]_{-T/2}^{T/2} = 0, \end{aligned}$$

because at $t = \pm T/2$ the arguments are integer multiples of π ($\omega T/2 = \pi$), and $\sin(k\pi) = 0$.

Case $m = n$:

$$\begin{aligned} \int_{-T/2}^{T/2} \sin^2(n\omega t) dt &= \frac{1}{2} \int_{-T/2}^{T/2} \cos(0) dt - \frac{1}{2} \int_{-T/2}^{T/2} \cos(2n\omega t) dt \\ &= \frac{1}{2} \left[t - \frac{\sin(2n\omega t)}{2n\omega} \right]_{-T/2}^{T/2} = \frac{1}{2} T = \frac{T}{2}. \end{aligned}$$

The cosine pair is identical with $\cos a \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b))$.

12 The Fourier series theorem

THEOREM Fourier series

A T -periodic function $f(t)$ that is absolutely integrable, with a finite number of discontinuities and finite jumps within one period, can be developed into a Fourier series

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t))$$

with coefficients

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt, \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega t) dt, \quad b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(n\omega t) dt.$$

READING THE THEOREM

- ♦ The basis $\mathcal{B} = \{1, \cos(n\omega t), \sin(n\omega t)\}_{n \geq 1}$ is an **orthogonal basis** for the space of square-integrable periodic functions of period $T = 2\pi/\omega$.
- ♦ The coefficients a_0, a_n, b_n are the **projections** of f onto those basis functions — each formula is an inner product of f with one basis element, divided by that element's "length squared".
- ♦ **Symmetry:** if f is even then $b_n = 0$ for all n ; if f is odd then $a_n = 0$ for all n .
- ♦ **At a jump** x^* , the series converges to the *average* of the one-sided limits: $\frac{1}{2}[f(x^{*+}) + f(x^{*-})]$.
- ♦ **Finite intervals:** any function defined on a finite interval may be *periodically extended* and then represented by a Fourier series.

INTUITION coefficients are projections

Because the basis is orthogonal, finding a_n is exactly like finding a vector's component along one axis: dot f with $\cos(n\omega t)$, and every *other* term integrates to zero, leaving only the piece you want. The factors $\frac{1}{T}$ and $\frac{2}{T}$ are just the normalisations $1/\int 1$ and $1/\int \cos^2$.

THEOREM Dirichlet conditions — when the series converges

If, over one period, f is (i) absolutely integrable, (ii) has only finitely many maxima and minima, and (iii) has only finitely many finite discontinuities, then its Fourier series **converges**: to $f(t)$ wherever f is continuous, and to the midpoint $\frac{1}{2}[f(t^{*-}) + f(t^{*+})]$ at each jump. These are *sufficient* conditions, and every signal in this course satisfies them.

13 Worked Fourier series

WORKED EXAMPLE 3 · triangular wave $f(t) = |t|$, period $T = 4$

Here $\omega = \frac{2\pi}{T} = \frac{\pi}{2}$, and f is **even**, so $b_n = 0$.

$$a_0 = \frac{1}{4} \int_{-2}^2 |t| dt = \frac{2}{4} \int_0^2 t dt = 1.$$

For a_n , integrate by parts (even symmetry doubles the half-range integral):

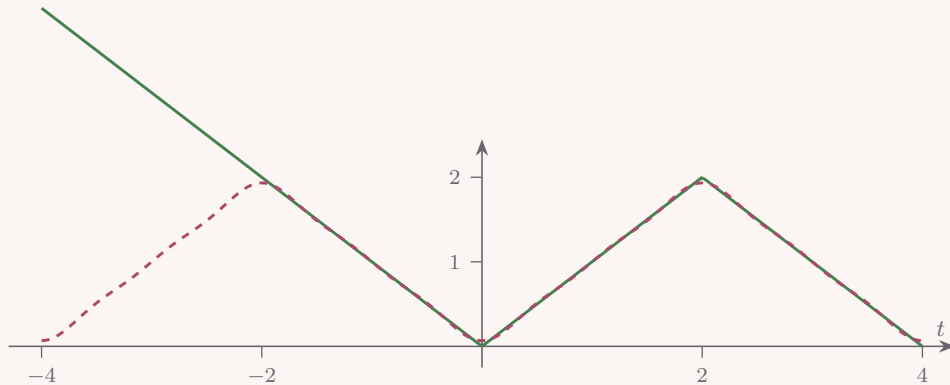
$$\begin{aligned} a_n &= \frac{2}{4} \int_{-2}^2 |t| \cos(n\omega t) dt = \frac{4}{4} \int_0^2 t \cos\left(\frac{n\pi}{2}t\right) dt = \frac{4}{n^2\pi^2} \left[\cos\left(\frac{n\pi t}{2}\right) + \frac{t}{2}n\pi \sin\left(\frac{n\pi t}{2}\right) \right]_0^2 \\ &= \frac{4}{n^2\pi^2} (\cos(n\pi) + n\pi \sin(n\pi) - \cos 0) = \frac{4}{n^2\pi^2} ((-1)^n - 1). \end{aligned}$$

Therefore, for $t \in [-2, 2]$,

$$f(t) = |t| = 1 + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} ((-1)^n - 1) \cos\left(n\frac{\pi}{2}t\right).$$

Two free identities by evaluating the series at special points:

$$f(0) = 0 : \quad \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} = -\frac{\pi^2}{4}, \quad f(1) = 1 : \quad \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(n\frac{\pi}{2}\right) = 0.$$



Triangular wave $f(t) = |t|$ (solid) and its three-term Fourier sum (dashed) — already a near-perfect fit, because the coefficients decay like $1/n^2$.

WORKED EXAMPLE 4 · sawtooth $f(t) = t$, period $T = 4$

Again $\omega = \frac{\pi}{2}$, but now f is **odd**, so $a_0 = a_n = 0$. Integrating by parts,

$$\begin{aligned} b_n &= \frac{2}{4} \int_{-2}^2 t \sin(n\omega t) dt = \frac{4}{4} \int_0^2 t \sin\left(\frac{n\pi}{2}t\right) dt = \frac{4}{n^2\pi^2} \left[\sin\left(\frac{n\pi t}{2}\right) - \frac{t}{2}n\pi \cos\left(\frac{n\pi t}{2}\right) \right]_0^2 \\ &= \frac{4}{n^2\pi^2} (\sin(n\pi) - n\pi \cos(n\pi)) = \frac{4}{n\pi} (-1)^{n+1}. \end{aligned}$$

Hence, for $t \in [-2, 2]$,

$$f(t) = t = \sum_{n=1}^{\infty} \frac{4}{n\pi} (-1)^{n+1} \sin\left(n\frac{\pi}{2}t\right).$$

Evaluating at $t = 1$ recovers the **Leibniz series**:

$$f(1) = 1 : \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(n\frac{\pi}{2}\right) = \frac{\pi}{4} \quad \left(= 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

INTUITION building a wave from harmonics

Each term $\sin(n\frac{\pi}{2}t)$ is a pure tone. Adding the fundamental and its harmonics with the right weights b_n gradually sculpts the straight ramp of the sawtooth — low harmonics give the gross shape, high harmonics sharpen the corners.

14 Half-range expansions: sine and cosine series

A function given only on $[0, L]$ can be extended to a periodic function in two natural ways, each killing half the coefficients. With $\omega = \pi/L$ (period $2L$):

DEFINITION Cosine and sine half-range series

Even (cosine) extension:

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{L}\right), \quad a_0 = \frac{1}{L} \int_0^L f \, dt, \quad a_n = \frac{2}{L} \int_0^L f \cos\left(\frac{n\pi t}{L}\right) dt.$$

Odd (sine) extension:

$$f(t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{L}\right), \quad b_n = \frac{2}{L} \int_0^L f \sin\left(\frac{n\pi t}{L}\right) dt.$$

INTUITION choose the symmetry you need

If the physics needs a *flat-ended* function (zero slope at the ends), reach for the cosine series; if it must *vanish at the ends*, reach for the sine series. This is exactly how Fourier series solve boundary-value problems for the heat and wave equations.

WORKED EXAMPLE · sine series of $f(t) = 1$ on $[0, L]$

$$b_n = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi t}{L}\right) dt = \frac{2}{n\pi} (1 - \cos n\pi) = \frac{2}{n\pi} (1 - (-1)^n),$$

which is $\frac{4}{n\pi}$ for odd n and 0 for even n . Hence, on $(0, L)$,

$$1 = \sum_{n \text{ odd}} \frac{4}{n\pi} \sin\left(\frac{n\pi t}{L}\right) = \frac{4}{\pi} \left(\sin \frac{\pi t}{L} + \frac{1}{3} \sin \frac{3\pi t}{L} + \frac{1}{5} \sin \frac{5\pi t}{L} + \dots \right).$$

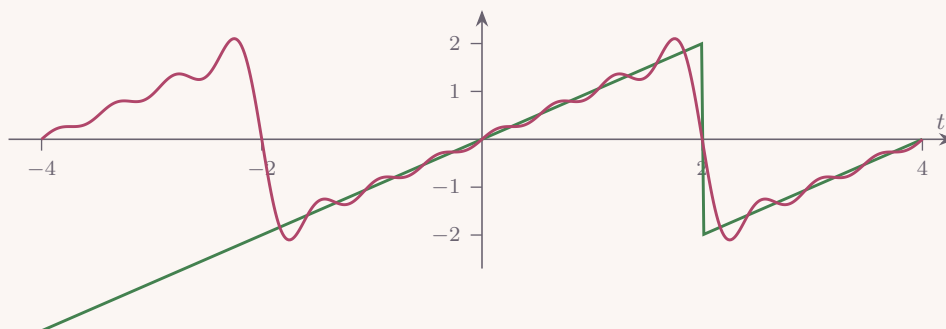
15 The Gibbs phenomenon

DEFINITION Gibbs phenomenon

The **Gibbs phenomenon** is the oscillatory behaviour that appears near a *jump discontinuity* when a discontinuous function is approximated by a *truncated* (finite) Fourier series.

BEHAVIOUR AT A DISCONTINUITY t^*

- ◆ At t^* the Fourier series equals the average $\frac{1}{2}(f(t^{*-}) + f(t^{*+}))$.
- ◆ Near t^* the partial sums **overshoot** above $f(t^{*+})$ and **undershoot** below $f(t^{*-})$.
- ◆ For a jump of amplitude $\Delta = |f(t^{*+}) - f(t^{*-})|$, the limiting overshoot (as $N \rightarrow \infty$) is about 9% of Δ .
- ◆ Adding more terms makes the oscillations *narrower*, but the maximum overshoot does **not** disappear.



Sawtooth $f(t) = t$ (solid) and a seven-term Fourier sum (rose): the little spikes near each jump are the Gibbs overshoot — they narrow as terms are added, but their height does not vanish.

16 Worked example: $f(t) = e^t$, period $T = \pi$

Here the circular frequency is $\omega = \frac{2\pi}{T} = 2$.

A NOTE ON THE SOURCE SLIDE

On the original lecture slide the worked integration for this example was hidden behind solid reveal-boxes, so the intermediate steps could not be read off. The coefficients below

are the standard result, reconstructed directly from the definitions; the lecturer's 27-term plot matches them.

Using the antiderivatives

$$\int e^t \cos(2nt) dt = \frac{e^t (\cos 2nt + 2n \sin 2nt)}{1 + 4n^2}, \quad \int e^t \sin(2nt) dt = \frac{e^t (\sin 2nt - 2n \cos 2nt)}{1 + 4n^2},$$

and evaluating over $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (so that $\cos(n\pi) = (-1)^n$, $\sin(n\pi) = 0$, and $e^{\pi/2} - e^{-\pi/2} = 2 \sinh \frac{\pi}{2}$):

$$a_0 = \frac{2}{\pi} \sinh \frac{\pi}{2}, \quad a_n = \frac{4 \sinh(\pi/2)}{\pi} \cdot \frac{(-1)^n}{1 + 4n^2}, \quad b_n = -\frac{8n \sinh(\pi/2)}{\pi} \cdot \frac{(-1)^n}{1 + 4n^2}.$$

So the Fourier series of the (period- π extension of the) exponential is

$$e^t = \frac{2}{\pi} \sinh \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{4 \sinh(\pi/2)}{\pi} \frac{(-1)^n}{1 + 4n^2} (\cos 2nt - 2n \sin 2nt).$$

The jumps at the ends of each period produce visible Gibbs overshoot in any truncation.

17 Decay rate and convergence

THEOREM Decay rate of the coefficients

If f is k times continuously differentiable (class $\mathcal{C}^k$), then its Fourier coefficients are bounded by

$$a_n = \mathcal{O}\left(\frac{1}{n^{k+1}}\right), \quad b_n = \mathcal{O}\left(\frac{1}{n^{k+1}}\right) \quad (n \rightarrow \infty).$$

More generally, if k is the lowest-order derivative that has a jump discontinuity, the same bound $\mathcal{O}(1/n^{k+1})$ holds. **Faster decay $\Rightarrow$ faster convergence**, i.e. fewer terms for a given accuracy.

INTUITION smoothness is paid back as speed

The smoother the signal, the more its energy sits in the low harmonics, so the coefficients die off quickly and a short sum already looks perfect. A signal with a corner or a jump leaks energy into high harmonics, and you must keep many terms.

Truncation error. In practice (audio compression, image processing, solving differential equations) we cannot sum infinitely many terms — we stop at some N (*truncation*). The error you incur is tied directly to the decay rate:

Function	Coeff. decay	Error after N	Terms needed
piecewise continuous (jump)	$1/n$	$\sim 1/N$	very slow — thousands of terms for a clean signal.
continuous, non-differentiable $\mathcal{C}^0$	$1/n^2$	$\sim 1/N^2$	much faster — a few dozen terms usually suffice.
very smooth $\mathcal{C}^4$	$1/n^5$	$\sim 1/N^5$	a handful of terms gives an almost perfect match.

18 Amplitude–phase form

The sine–cosine pair at each frequency can be merged into a single shifted cosine.

DEFINITION Amplitude–phase (harmonic) form

$$f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega t - \varphi_n), \quad A_n = \sqrt{a_n^2 + b_n^2}, \quad \varphi_n = \arctan\left(\frac{b_n}{a_n}\right).$$

A_n is the **amplitude** of the n -th harmonic and φ_n its **phase**.

CHOOSING THE PHASE QUADRANT

The phase is taken in $\varphi \in]-\pi, \pi]$, and — just as with Arg — the signs of a_n (the "x") and b_n (the "y") fix the quadrant:

$$\begin{aligned} \varphi_1 &= \arctan\left(\frac{2}{0}\right) = \frac{\pi}{2}, & \varphi_2 &= \arctan\left(\frac{-2}{0}\right) = -\frac{\pi}{2}, \\ \varphi_3 &= \arctan\left(\frac{0}{2}\right) = 0, & \varphi_4 &= \arctan\left(\frac{0}{-2}\right) = \pi. \end{aligned}$$

19 Spectra: time vs. frequency domain

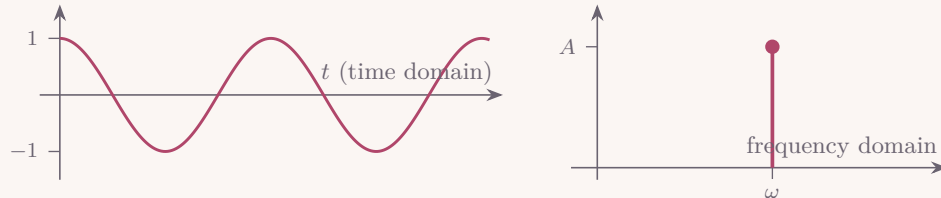
INTUITION two views of one signal

A single tone $x(t) = A \cos(\omega t)$ is a wave drawn against *time* (period $T = 2\pi/\omega = 1/\text{freq}$) — the **time domain**. The very same tone is *one spike* of height A at frequency ω when drawn against *frequency* — the **frequency domain**. Fourier analysis is the dictionary between these two pictures.

DEFINITION Amplitude spectrum

The spectrum of a periodic signal is **discrete**: a set of spikes at the frequencies $0, \omega, 2\omega, 3\omega, \dots$, consisting of

- ♦ a **DC component** a_0 (the constant / mean value) at frequency 0;
- ♦ the **fundamental** at ω with amplitude A_1 ;
- ♦ the **harmonics** at the integer multiples $n\omega$ with amplitudes A_n .



A pure tone $x(t) = A \cos(\omega t)$ is one wave drawn against time (left), and a single spectral line at ω drawn against frequency (right).

WORKED EXAMPLE 6 · spectrum of the triangular wave $f(t) = |t|$ (from Ex. 3)

With $a_0 = 1$, $a_n = \frac{4}{n^2\pi^2}((-1)^n - 1)$, $b_n = 0$:

$$A_n = \sqrt{a_n^2 + b_n^2} = \frac{4}{n^2\pi^2} |(-1)^n - 1|, \quad \varphi_n = \arctan\left(\frac{0}{a_n}\right).$$

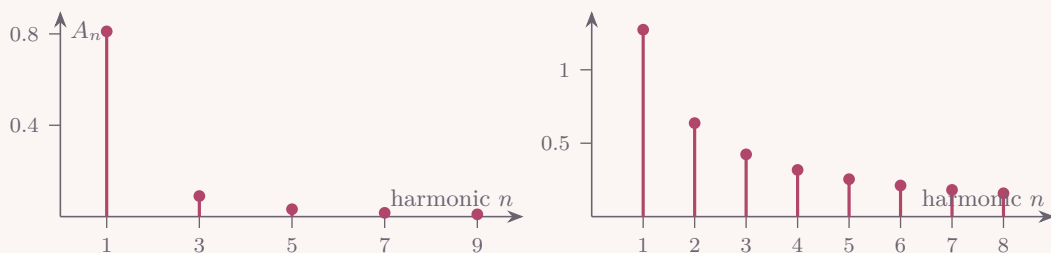
Since $(-1)^n - 1 = 0$ for even n , *only the odd harmonics survive*; here $\omega = \frac{\pi}{2}$ rad/s so $\text{freq} = \frac{\omega}{2\pi} = 0.25$ Hz. Because $a_n < 0$ on the odd harmonics, their phase is π (and 0 elsewhere).

WORKED EXAMPLE 7 · spectrum of the sawtooth $f(t) = t$ (from Ex. 4)

With $a_0 = a_n = 0$ and $b_n = \frac{4}{n\pi}(-1)^{n+1}$:

$$A_n = \frac{4}{n\pi} \quad (\text{a slow } 1/n \text{ decay, every harmonic present}), \quad \varphi_n = \begin{cases} \frac{\pi}{2}, & n \text{ odd} \\ -\frac{\pi}{2}, & n \text{ even.} \end{cases}$$

Equivalently, in harmonic form, $t = \sum_{n=1}^{\infty} \frac{4}{n\pi} \cos\left(n\frac{\pi}{2}t + (-1)^n\frac{\pi}{2}\right)$.



Amplitude spectra. Left: the triangular wave $|t|$ — only odd harmonics, dying like $1/n^2$. Right: the sawtooth t — every harmonic present, dying only like $1/n$ (hence the slow convergence and Gibbs ringing).

20 Complex form of the Fourier series

Substituting Euler's identities $\cos(n\omega t) = \frac{1}{2}(e^{in\omega t} + e^{-in\omega t})$ and $\sin(n\omega t) = \frac{1}{2i}(e^{in\omega t} - e^{-in\omega t})$ into the real series and collecting the $e^{\pm in\omega t}$ terms gives the complex coefficients

$$c_n = \frac{a_n - ib_n}{2}, \quad c_{-n} = \bar{c}_n = \frac{a_n + ib_n}{2}, \quad c_0 = a_0, \quad |c_n| = |c_{-n}|.$$

THEOREM Complex Fourier series

The basis in complex form is the set of orthogonal complex exponentials $\{e^{in\omega t} \mid n \in \mathbb{Z}\}$, and

$$f(t) = \sum_{n=-\infty}^{+\infty} c_n e^{in\omega t}, \quad c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-in\omega t} dt.$$

The single complex formula packs the cosine and sine integrals together, since $c_n = \frac{1}{T} \int f(t)(\cos(n\omega t) - i \sin(n\omega t)) dt$.

WORKED EXAMPLE 8 · square wave

Let $f(t) = \begin{cases} -1, & t \in [-1, 0) \\ 1, & t \in [0, 1) \end{cases}$ with period $T = 2$, so $\omega = \frac{2\pi}{T} = \pi$. Then

$$\begin{aligned} c_n &= \frac{1}{2} \int_{-1}^1 f(t) e^{-in\pi t} dt = \frac{1}{2} \int_{-1}^0 (-1) e^{-in\pi t} dt + \frac{1}{2} \int_0^1 (1) e^{-in\pi t} dt \\ &= \frac{1}{2} \left[\frac{e^{-in\pi t}}{-in\pi} \right]_{-1}^0 + \frac{1}{2} \left[-\frac{e^{-in\pi t}}{in\pi} \right]_0^1 = \frac{1}{2in\pi} (2 - e^{in\pi} - e^{-in\pi}). \end{aligned}$$

Since $e^{in\pi} = e^{-in\pi} = (-1)^n$ for all $n \in \mathbb{Z}$,

$$c_n = \frac{1}{in\pi} (1 - (-1)^n) = \frac{i}{n\pi} ((-1)^n - 1), \quad f(t) = \sum_{n=-\infty}^{+\infty} \frac{i}{n\pi} ((-1)^n - 1) e^{in\pi t}.$$

Only *odd* n contribute ($c_n = -\frac{2i}{n\pi}$ for odd n , and $c_0 = 0$ since the mean is zero).

21 Operational properties

These rules transform one Fourier series into another without recomputing any integral. Stated for the complex coefficients c_n of a T -periodic signal ($\omega = 2\pi/T$):

COEFFICIENT RULES

- ◆ **Linearity:** $\alpha f + \beta g \longleftrightarrow \alpha c_n^f + \beta c_n^g$.
- ◆ **Time shift:** $f(t - t_0) \longleftrightarrow c_n e^{-in\omega t_0}$ — the amplitudes $|c_n|$ are unchanged; only the

phases rotate.

- ♦ **Differentiation:** $f'(t) \longleftrightarrow i n \omega c_n$ — differentiating multiplies harmonic n by $i n \omega$, which is precisely why rougher (less differentiable) signals have slower-decaying coefficients.
- ♦ **Real signals:** f is real $\iff c_{-n} = \bar{c}_n$ — the spectrum is conjugate-symmetric.

22 The three forms at a glance

The same periodic signal has three equivalent Fourier representations; use whichever suits the task.

Form	Series and coefficients
Trigonometric	$f = a_0 + \sum_{n \geq 1} (a_n \cos n\omega t + b_n \sin n\omega t)$
Amplitude-phase	$f = a_0 + \sum_{n \geq 1} A_n \cos(n\omega t - \varphi_n), \quad A_n = \sqrt{a_n^2 + b_n^2}, \quad \varphi_n = \arctan \frac{b_n}{a_n}$
Complex exponential	$f = \sum_{n=-\infty}^{\infty} c_n e^{i n \omega t}, \quad c_n = \frac{1}{2}(a_n - i b_n), \quad c_0 = a_0, \quad c_{-n} = \bar{c}_n$

The bridge: $c_0 = a_0$, and for $n \geq 1$, $A_n = 2|c_n| = \sqrt{a_n^2 + b_n^2}$, with the phase φ_n carried by $\arg(c_n)$.

23 Energy, power, and Parseval

DEFINITION Energy and power

The **energy** of a signal is $E = \int_{-\infty}^{+\infty} |f(t)|^2 dt$. For a periodic signal this diverges, so a periodic signal has **infinite energy**. The right measure is its **power** — the average energy per period:

$$\mathcal{P} = \lim_{n \rightarrow \infty} \frac{1}{nT} \int_{-nT/2}^{nT/2} |f(t)|^2 dt = \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt = \frac{E_{\text{period}}}{T} = \bar{\mathcal{P}}.$$

THEOREM Parseval — conservation of power

The power computed in the time domain equals the power computed in the frequency domain:

$$\bar{\mathcal{P}} = \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \sum_{n=-\infty}^{+\infty} |c_n|^2.$$

INTUITION slice by time, or by harmonic

Whether you add up the signal slice-by-slice over one period, or harmonic-by-harmonic over frequency, the total "weight" is identical. Parseval is the statement that the change of basis to $\{e^{i\omega t}\}$ preserves length — energy is merely re-bookkept across frequencies, never created or lost.

PART III**The Fourier Transform** フーリエ変換**24 From series to transform**

A Fourier *series* builds a periodic signal from a discrete comb of harmonics $n\omega$. The Fourier *transform* is what happens when the period grows without bound.

DEFINITION Fourier transform and its inverse

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega t} d\omega.$$

$\mathcal{F}$ carries a signal from t -space to ω -space; $\mathcal{F}^{-1}$ brings it back.

INTUITION from a line spectrum to a continuous one

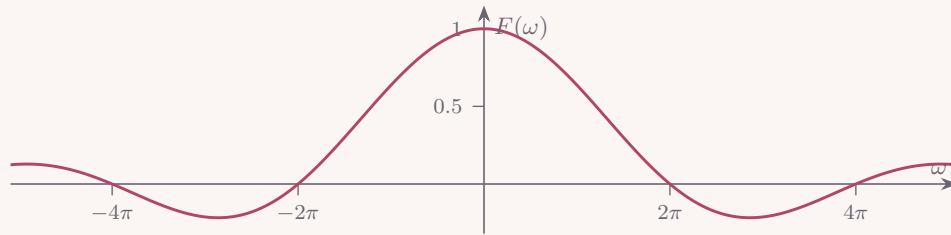
As the period $T \rightarrow \infty$ the harmonic spacing $\Delta\omega = 2\pi/T \rightarrow 0$, so the discrete spectrum of a Fourier series fills in to a *continuous* curve and the sum becomes an integral. A periodic signal has energy only at the discrete $n\omega$; a one-off pulse spreads its content over *all* frequencies, and $|F(\omega)|$ measures how much of each it contains.

25 Two fundamental transforms**WORKED EXAMPLE 1** · rectangular pulse and the sinc

Let $f(t) = \begin{cases} 1, & |t| < \frac{1}{2} \\ 0, & |t| \geq \frac{1}{2}. \end{cases}$ Then

$$F(\omega) = \int_{-1/2}^{1/2} e^{-i\omega t} dt = \left[-\frac{e^{-i\omega t}}{i\omega} \right]_{-1/2}^{1/2} = \frac{e^{i\omega/2} - e^{-i\omega/2}}{i\omega} = \frac{\sin(\omega/2)}{\omega/2} = \text{sinc}\left(\frac{\omega}{2}\right).$$

A sharp gate in time becomes a slowly-decaying sinc in frequency: the narrower the pulse, the wider its spectrum.



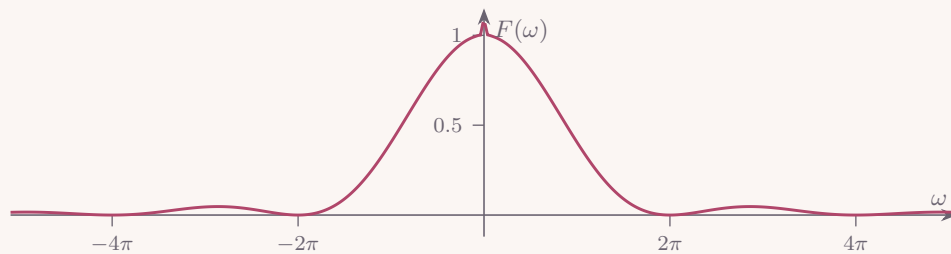
Spectrum of the unit pulse: $F(\omega) = \text{sinc}(\omega/2) = \sin(\omega/2)/(\omega/2)$. A gate in time spreads into a slowly decaying sinc in frequency.

WORKED EXAMPLE 2 · the triangle function

Let $f(t) = \begin{cases} 1 - |t|, & |t| \leq 1 \\ 0, & \text{otherwise.} \end{cases}$ Splitting the integral at 0,

$$F(\omega) = \int_{-1}^0 (1+t)e^{-i\omega t} dt + \int_0^1 (1-t)e^{-i\omega t} dt = \frac{2(1 - \cos \omega)}{\omega^2}.$$

The triangle is the convolution of two pulses, so its transform is essentially the *square* of a sinc — everywhere ≥ 0 and decaying like $1/\omega^2$.



Spectrum of the triangle: $F(\omega) = 2(1 - \cos \omega)/\omega^2 = \text{sinc}^2(\omega/2)$ — everywhere non-negative and decaying like $1/\omega^2$, the square of the pulse's sinc.

26 Existence

DEFINITION The space $L^1(\mathbb{R})$

$f \in L^1(\mathbb{R})$ means f is **absolutely integrable**:

$$\int_{-\infty}^{+\infty} |f(t)| dt < \infty.$$

If $f \in L^1(\mathbb{R})$, its Fourier transform is guaranteed to exist.

27 Properties

FOURIER TRANSFORM RULES

- ♦ **Linearity:** $\mathcal{F}\{af + bg\} = a\mathcal{F}\{f\} + b\mathcal{F}\{g\}$.
- ♦ **Time shift:** $\mathcal{F}\{f(t - t_0)\} = F(\omega) e^{-i\omega t_0}$ — a delay leaves $|F|$ unchanged and adds a linear phase.
- ♦ **Frequency shift (modulation):** $\mathcal{F}\{f(t)e^{i\omega_0 t}\} = F(\omega - \omega_0)$ — multiplying by a complex exponential slides the whole spectrum.
- ♦ **Time scaling:** $\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$ — compress in time and you stretch in frequency.
- ♦ **Differentiation:** $\mathcal{F}\{f'(t)\} = i\omega F(\omega)$ — each time-derivative becomes multiplication by $i\omega$.

THEOREM Parseval — energy conservation

$$\int_{-\infty}^{+\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |F(\omega)|^2 d\omega.$$

The quantity $|F(\omega)|^2$ is the **energy spectrum** — energy per unit frequency.

INTUITION why differentiation becomes multiplication matters

Since $\frac{d}{dt} \rightarrow i\omega$, a *differential* equation in t turns into an *algebraic* equation in ω : solve the algebra, then invert. This is the whole reason transforms are the engineer's tool for ODEs and PDEs.

WORKED EXAMPLE 3 · the one-sided decaying exponential

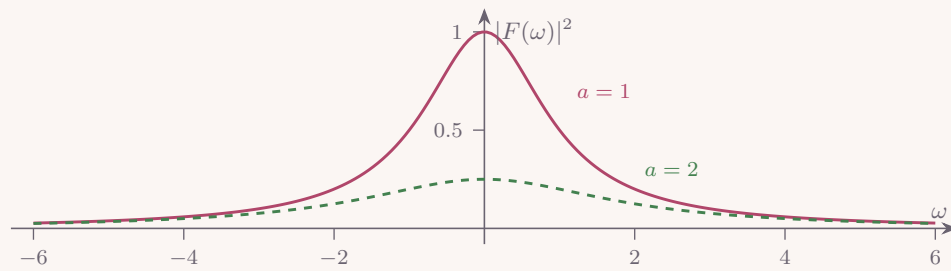
Let $f(t) = \begin{cases} 0, & t \leq 0 \\ e^{-at}, & t > 0 \end{cases}$ with $a > 0$. Then

$$F(\omega) = \int_0^{+\infty} e^{-at} e^{-i\omega t} dt = \int_0^{+\infty} e^{-(a+i\omega)t} dt = \left[-\frac{e^{-(a+i\omega)t}}{a+i\omega} \right]_0^{+\infty} = \frac{1}{a+i\omega},$$

a *complex* spectrum, whose energy spectrum is the Lorentzian

$$|F(\omega)|^2 = \frac{1}{a^2 + \omega^2}.$$

A larger a (faster decay in time) gives a wider, flatter spectrum.



Energy spectrum $|F(\omega)|^2 = 1/(a^2 + \omega^2)$ of the one-sided exponential. Faster decay in time (larger a) gives a wider, lower, flatter spectrum.

28 When is the spectrum real?

Split the kernel with $e^{-i\omega t} = \cos \omega t - i \sin \omega t$:

$$F(\omega) = \underbrace{\int_{-\infty}^{+\infty} f(t) \cos(\omega t) dt}_{\text{real part}} - i \underbrace{\int_{-\infty}^{+\infty} f(t) \sin(\omega t) dt}_{\text{imaginary part}}.$$

SYMMETRY OF REAL SIGNALS

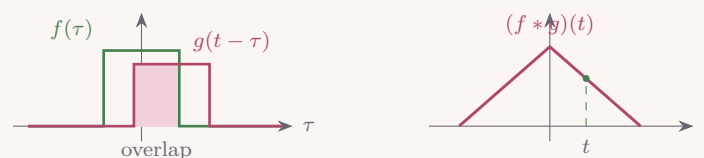
- ♦ If f is real and **even**, then $F(\omega)$ is real and even.
- ♦ If f is real and **odd**, then $F(\omega)$ is purely imaginary and odd.

29 Convolution

Convolution is the operation that sits underneath both transforms — it is how one signal is blended with, or "smeared" by, another.

DEFINITION Convolution

$$(f * g)(t) = \int_{-\infty}^{+\infty} f(\tau) g(t - \tau) d\tau.$$



Convolution as flip-and-slide. Left: for each shift t , reflect g , slide it by t , and integrate the shaded overlap with f . Right: doing this for two unit pulses traces a triangle — so pulse $*$ pulse = triangle.

INTUITION flip, slide, integrate the overlap

To evaluate $(f * g)$ at a shift t : reflect $g(\tau) \mapsto g(-\tau)$, slide it to $g(t - \tau)$, multiply pointwise by $f(\tau)$, and integrate. The result is the *area of overlap* between f and the flipped, shifted

g .

ALGEBRA OF CONVOLUTION

- ♦ Commutative $f * g = g * f$; associative $(f * g) * h = f * (g * h)$; distributive over $+$.
- ♦ The Dirac delta is the identity: $f * \delta = f$.
- ♦ It differentiates cleanly: $(f * g)' = f' * g = f * g'$.

THEOREM Convolution theorem

Convolution in one domain is multiplication in the other:

$$\mathcal{F}\{f * g\} = F(\omega) G(\omega), \quad \mathcal{F}\{f g\} = \frac{1}{2\pi} (F * G)(\omega).$$

INTUITION why filtering is multiplication in frequency

Passing a signal through a linear, time-invariant **filter** *convolves* it with the filter's impulse response in time — which, by the theorem, is the same as *multiplying* its spectrum by the filter's frequency response. Designing a filter is then just choosing which frequencies to keep and which to suppress.

WORKED EXAMPLE · pulse * pulse = triangle

Let $p(t) = \begin{cases} 1, & |t| < \frac{1}{2} \\ 0, & \text{otherwise.} \end{cases}$ As one copy slides across the other, the overlap grows then shrinks linearly, giving

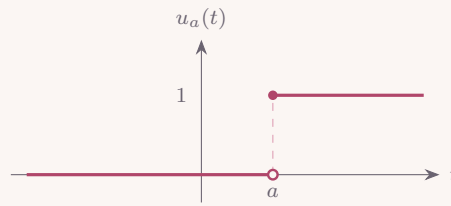
$$(p * p)(t) = \begin{cases} 1 - |t|, & |t| \leq 1 \\ 0, & \text{otherwise,} \end{cases}$$

exactly the triangle of Example 2. The convolution theorem confirms it: $\mathcal{F}\{p * p\} = \mathcal{F}\{p\}^2 = \text{sinc}^2(\omega/2)$, and indeed $2(1 - \cos \omega)/\omega^2 = \text{sinc}^2(\omega/2)$ because $1 - \cos \omega = 2 \sin^2(\omega/2)$.

30 The unit step, and piecewise signals

DEFINITION Unit (Heaviside) step

$$u_a(t) = u(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a. \end{cases}$$



The unit (Heaviside) step $u_a(t)$ switches from 0 to 1 at $t = a$.

Any piecewise signal is assembled from steps — switch a piece *on* at its start and *off* at its end. If f follows laws f_1, f_2, f_3, f_4 on successive intervals split at a, b, c ,

$$f(t) = f_1(1 - u_a(t)) + f_2(u_a(t) - u_b(t)) + f_3(u_b(t) - u_c(t)) + f_4 u_c(t).$$

WORKED EXAMPLE 4 · a four-piece signal

For the signal equal to $3e^t$ on $(-\infty, 1)$, $\ln t$ on $[1, 3)$, t on $[3, 5)$, and -3 on $[5, \infty)$,

$$f(t) = 3e^t(1 - u_1(t)) + \ln t(u_1(t) - u_3(t)) + t(u_3(t) - u_5(t)) - 3u_5(t),$$

which regroups into the compact "jump" form

$$f(t) = 3e^t + u_1(t)(\ln t - 3e^t) + u_3(t)(t - \ln t) + u_5(t)(-3 - t).$$

THE BRIDGE TO LAPLACE

This step-function bookkeeping is exactly what makes the Laplace *shifting theorem* so powerful for signals that switch on at a given time — the subject of Part IV.

PART IV

The Laplace Transform

ラプラス変換

31 Integral transforms, and the Laplace definition

Both the Fourier and Laplace transforms are **integral transforms**: multiply the signal by a kernel $K(s, t)$ and integrate.

DEFINITION Integral transform

$$\mathfrak{I}\{f(t)\} = \int_{\alpha}^{\beta} f(t) K(s, t) dt,$$

fixed by three ingredients: the input $f(t)$, the **kernel** $K(s, t)$ (the bridge from t to the new variable s), and the limits $[\alpha, \beta]$.

Two choices of kernel give the transforms of this course:

Fourier: $K = e^{-i\omega t}$, $[\alpha, \beta] = (-\infty, +\infty)$; Laplace: $K = e^{-st}$, $[\alpha, \beta] = [0, +\infty)$.

DEFINITION Laplace transform

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{+\infty} f(t) e^{-st} dt.$$

INTUITION Laplace = Fourier with a damping factor

Write $s = \sigma + i\omega$, so $e^{-st} = e^{-\sigma t} e^{-i\omega t}$ — a Fourier kernel times a decaying envelope $e^{-\sigma t}$. That envelope lets the integral converge for growing signals the Fourier transform cannot handle, and starting the integral at $t = 0$ makes Laplace the natural tool for *initial-value* problems.

32 Existence

DEFINITION Exponential order

The Laplace transform exists when the integral converges — for functions that do not grow faster than an exponential (are of **exponential order**) and have no singularities on $[0, +\infty)$.

TRANSFORMS THAT DO NOT EXIST

e^{t^2} grows faster than any exponential, and $1/t$ has a singularity at $t = 0$; neither has a Laplace transform.

WORKED EXAMPLE 5 · the exponential $\mathcal{L}\{e^{at}\}$

$$\mathcal{L}\{e^{at}\} = \int_0^{+\infty} e^{at} e^{-st} dt = \int_0^{+\infty} e^{(a-s)t} dt = \left[\frac{e^{(a-s)t}}{a-s} \right]_0^{+\infty} = \frac{1}{s-a},$$

valid for $s > a$, so that $e^{(a-s)t} \rightarrow 0$ at the upper limit.

33 Table of transforms

Throughout, a, b are constants, $n \in \mathbb{N}$, and u_a, δ_a are the unit step and the Dirac impulse at $t = a$.
Elementary functions.

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
$\sin(bt)$	$\frac{b}{s^2 + b^2}$
$\cos(bt)$	$\frac{s}{s^2 + b^2}$
$\sinh(bt)$	$\frac{b}{s^2 - b^2}$
$\cosh(bt)$	$\frac{s}{s^2 - b^2}$
$t^n \ (n \in \mathbb{N})$	$\frac{n!}{s^{n+1}}$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$

Resonant and damped pairs.

$f(t)$	$F(s)$
$t \sin(bt)$	$\frac{2bs}{(s^2 + b^2)^2}$
$t \cos(bt)$	$\frac{s^2 - b^2}{(s^2 + b^2)^2}$
$e^{-at} \sin(bt)$	$\frac{b}{(s+a)^2 + b^2}$
$e^{-at} \cos(bt)$	$\frac{s+a}{(s+a)^2 + b^2}$
$\sin(bt) - bt \cos(bt)$	$\frac{2b^3}{(s^2 + b^2)^2}$
$\frac{t \sin(bt)}{2b}$	$\frac{s}{(s^2 + b^2)^2}$

Steps, impulses, and operational rules.

$f(t)$	$F(s)$
$u_a(t)$	$\frac{e^{-as}}{s}$
$u_a(t) f(t-a)$	$e^{-as} F(s)$
$u_a(t) f(t)$	$e^{-as} \mathcal{L}\{f(t+a)\}$
$\delta_a(t)$	e^{-as}
$e^{-at} f(t)$	$F(s+a)$
$\int_0^t f(\zeta) d\zeta$	$\frac{F(s)}{s}$

34 Shifting theorems, impulses, and integration

THEOREM The two shifting theorems

First shift (in s): multiplying by e^{-at} shifts the transform,

$$\mathcal{L}\{e^{-at}f(t)\} = F(s+a).$$

Second shift (in t): delaying a signal by a and switching it on with u_a ,

$$\mathcal{L}\{u_a(t)f(t-a)\} = e^{-as}F(s).$$

Alongside these: the Dirac impulse transforms to $\mathcal{L}\{\delta_a(t)\} = e^{-as}$, and integrating in time divides by s , $\mathcal{L}\{\int_0^t f(\zeta) d\zeta\} = F(s)/s$.

35 The derivative theorem

This is the property that lets Laplace solve differential equations. Integrating by parts,

$$\int_0^\infty e^{-st}f'(t) dt = [e^{-st}f(t)]_0^\infty - \int_0^\infty f(t)(-se^{-st}) dt = sF(s) - f(0).$$

THEOREM Transform of derivatives

$$\begin{aligned}\mathcal{L}\{f'(t)\} &= sF(s) - f(0), \\ \mathcal{L}\{f''(t)\} &= s^2F(s) - sf'(0) - f''(0), \\ \mathcal{L}\{f'''(t)\} &= s^3F(s) - s^2f'(0) - sf''(0) - f'''(0), \\ \mathcal{L}\{f^{(n)}(t)\} &= s^nF(s) - s^{n-1}f'(0) - s^{n-2}f''(0) - \dots - f^{(n-1)}(0).\end{aligned}$$

INTUITION derivatives carry the initial conditions

Each derivative brings down a power of s and subtracts the initial data $f(0), f'(0), \dots$. The Laplace transform therefore folds the initial conditions straight into the algebra — there are no arbitrary constants left to fix at the end.

WORKED EXAMPLE 8 · $\mathcal{L}\{\sin^2 t\}$ via the derivative rule

With $f = \sin^2 t$ we have $f(0) = 0$ and $f'(t) = 2 \sin t \cos t = \sin 2t$. Then $\mathcal{L}\{f'\} = s\mathcal{L}\{f\} - f(0)$ gives

$$\mathcal{L}\{\sin 2t\} = s\mathcal{L}\{\sin^2 t\} \implies \mathcal{L}\{\sin^2 t\} = \frac{1}{s}\mathcal{L}\{\sin 2t\} = \frac{1}{s} \cdot \frac{2}{s^2 + 4} = \frac{2}{s(s^2 + 4)}.$$

36 Inverse transforms by partial fractions

INTUITION the strategy

To invert a rational $F(s)$, break it into simple pieces whose inverses appear in the table — usually by *partial fractions* — then read each piece off the table and add the results.

WORKED EXAMPLE 6 · inverse of $\frac{3}{s(s+1)(s+3)}$

Write

$$Y(s) = \frac{3}{s(s+1)(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3}, \quad A = 1, \quad B = -\frac{3}{2}, \quad C = \frac{1}{2}.$$

Using rows 1 and 2 of the table,

$$y(t) = \mathcal{L}^{-1}\{Y\} = 1 - \frac{3}{2}e^{-t} + \frac{1}{2}e^{-3t}.$$

WORKED EXAMPLE 7 · inverse of $\frac{e^{-3s}}{s^2}$

The factor e^{-3s} signals the second shifting theorem with $a = 3$ and $F(s) = 1/s^2$, i.e. $f(t) = t$. Hence

$$Y(s) = e^{-3s} \mathcal{L}\{t\} = \mathcal{L}\{u_3(t)(t-3)\} \implies y(t) = u_3(t)(t-3).$$

37 Convolution and the impulse response

For *causal* signals (zero for $t < 0$) the convolution integral runs over $[0, t]$:

$$(f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau.$$

THEOREM Convolution theorem (Laplace)

$$\mathcal{L}\{(f * g)(t)\} = F(s) G(s).$$

A *product* of transforms inverts to a *convolution* — invaluable when partial fractions get unwieldy.

WORKED EXAMPLE · an inverse transform by convolution

With $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ and $\mathcal{L}\{e^{bt}\} = \frac{1}{s-b}$ (and $a \neq b$),

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-a)(s-b)}\right\} = e^{at} * e^{bt} = \int_0^t e^{a\tau} e^{b(t-\tau)} d\tau = \frac{e^{at} - e^{bt}}{a - b}.$$

INTUITION the transfer function multiplies; time domain convolves

Model a linear system by its **impulse response** $h(t)$, whose transform $H(s) = \mathcal{L}\{h\}$ is the **transfer function**. The response to any input $x(t)$ is the convolution $y = h * x$, i.e. $Y(s) = H(s)X(s)$: in the s -domain the system merely *multiplies*, while back in time it *smears* the input by h . This is the gateway to the transfer-function methods of Lecture 5.

38 Solving ODEs with the Laplace transform

INTUITION turn calculus into algebra

Transform the entire equation: derivatives become powers of s (carrying the initial conditions), the ODE becomes an algebraic equation in $Y(s)$, you solve for $Y(s)$, and a table look-up — after partial fractions — returns $y(t)$.

WORKED EXAMPLE · solve $y'' + y = 1$, $y(0) = y_0$, $y'(0) = v_0$

Transforming termwise with the derivative theorem,

$$(s^2 + 1)Y - s y_0 - v_0 = \frac{1}{s} \implies Y = \frac{s y_0}{s^2 + 1} + \frac{v_0}{s^2 + 1} + \frac{1}{s(s^2 + 1)}.$$

Splitting the last term, $\frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1}$, so

$$Y = \mathcal{L}\{\cos t\} y_0 + \mathcal{L}\{\sin t\} v_0 + \mathcal{L}\{1 - \cos t\}.$$

Inverting term by term,

$$y(t) = y_0 \cos t + v_0 \sin t + (1 - \cos t).$$

The first two terms are the **natural response** (fixed by the initial state y_0, v_0); the last is the **forced response** to the constant input 1.



完 · end of notes · ASE203 Parts I–IV